CHARACTERIZING NON-IDEAL SHAPES IN TERMS OF DIMENSIONS AND TOLERANCES

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Abstract: A geometric model of a shape is extended so as to represent not only its nominal dimensions but also tolerance information and surface specifications. The data structure defining an object is visualised as a pin-jointed, infinitely elastic wire frame covered by elastic membranes. Constraints corresponding to the dimensions of technical drawings are applied to the structure, either by the designer or by an automatic dimensioning algorithm. The validity of the resultant scheme can be checked and then drawn using drawing office conventions. The data structure can handle dimensions fixing various types of curved surface; other types can easily be added by following the same rules. A method of changing the nominal shape of an object is demonstrated. These results provide a systematic account of the ill-defined art of dimensioning and tolerancing, and should make the draughting process more amenable to computation.

Keywords and Phrases: Computer-aided design, dimensioning, draughting, geometric modelling, graphics, mechanical components, tolerances.

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1. Introduction

The earliest engineering drawings evolved from pictures, more or less to scale. They were sufficient for a single craftsman to work from, and for some purposes, are still adequate today. As the volume of production increased, manufacture was divided into stages. For example, components were made separately, and later assembled. No doubt difficulties in assembly arose and the job of fitter was born. The same difficulties also led to the addition of dimensions to the pictures.

Now a new problem appeared - that of redundancy. Dimensions could be in conflict with the picture or with one another. The first conflict was solved by giving dimensions precedence over the picture. Later, it became customary to add enough dimensions to define all lengths and angles, leaving the picture to convey only the arrangement (or topology) of the shape. Stern warnings - "Do not scale" - were stamped on every drawing. Conflict between dimensions demanded a more subtle response. The rule was that the dimension scheme should be just sufficient to define the shape. One dimension more or less would be wrong, and even if the total number of dimensions were correct, an inappropriate choice could leave a shape over-defined in one region and under-defined elsewhere. The matter was further complicated by various short cuts that allowed 'obvious' dimensions like right-angles to be omitted, or capitalised on symmetry or patterns. Thus in practice, the formal rules were often disobeyed, especially in the case of non-trivial shapes. Sometimes the intelligence and experience of the reader could resolve conflicts or supply missing information but often mistakes were made.

For any given shape, there is a multitude of correct dimensioning schemes that differ only in their choice of lengths and angles to be given (i.e. dimensioned) and those to be derived. This extra information is put to good use as a means of stressing certain lengths and angles (those that are given) at the expense of others, thereby conveying to the maker or inspector of a shape those parts of the shape that are significant for assembly or function. The addition of tolerances adds further information by specifying how close a given length or angle should be to its nominal value. A problem that designers sometimes face is to find the effect of tolerances on a derived length or angle, and much tedious calculation may be needed.

Existing computer systems for handling shape information are mostly at the picture stage. They model shapes but give no precedence to particular lengths or angles, nor do they characterize precision explicitly. For many practical purposes, a linear analysis of dimensions is sufficient. The PADL system [3] applies such a technique to rectilinear shapes, treating each orthogonal axis independently. In general however the axes interact and engineering components are not rectilinear. Hence, a more elaborate analysis is required. In [2], the authors have shown how dimensions and tolerances can be added to a shape description, how a dimension scheme can be checked for incompleteness and redundancy, and how tolerance calculations can be automated. The

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results, for polyhedra, are summarised below. This method is reminiscent of the Sketchpad system [4] and its handling of constraints. Further possibilities will be explored in this paper. These are the automatic generation of dimensioning schemes, the treatment of curved surfaces, and the use of dimensions as a design variable, that is, as a means of changing stored shape descriptions in a natural and easily comprehended way.

We shall assume that the precision of the computer representation and operations is much finer than the tolerances being manipulated. But it may be observed in passing that problems of redundancy and consistency in computer models of shape are analogous and could perhaps benefit from a similar treatment.

2. Analysis of dimensions and tolerances

The underlying method in analysing dimensions and tolerances is to set up equations relating shape geometry, that is, the equations of curves and surfaces, and the positions of vertex points, to the dimensions shown or implied in the drawing. When the dimensions vary within their tolerances, the shape alters simultaneously. Because tolerances are small in relation to dimensions, we can neglect second and higher order small quantities and make the equations linear.

The variables of shape geometry must be chosen carefully so as to be independent. Two surfaces intersecting in a curve define the curve. If more than three surfaces meet in a vertex point, then only three of the surfaces are fully independent. A dimensioning scheme cannot settle this choice and any program to handle dimensions and tolerances must make a somewhat arbitrary decision.

In [2], the vertex points are chosen as the unknowns. Consequently, a planar face with more than three vertices implies that the vertices are not independent. Every dimension gives rise to one or more equations, and the fact that a correct number of dimensions is present can be established by checking that there are as many independent equations as unknowns. The right-hand side terms in the system of equations are the variations in any dimension as permitted by the tolerances. For a given set of variations, we can solve to find the vertex positions provided the equations are well-conditioned. If not, we conclude that the dimensioning scheme is poor and that the shape is ill-defined; at worst, it may be a mechanism, that is a non-rigid shape. In the well-conditioned case, we may also invert the equations to obtain vertex coordinates as a function of the dimension variations. This is precisely what is needed when analysing dimensioning and tolerance schemes. For example, we can find the resulting tolerance on an undimensioned length and moreover can discover the proportion contributed by each given tolerance. The system of equations can be represented in matrix form by $R\,d = u$ where R is called the rigidity matrix and u and d are vectors of the variations and vertex displacements respectively.

In practice we begin with a stored shape model made with a program such as BUILD [1]. The model must explicitly contain the shape topology; that is, all faces, edges and vertices must be present, together with details of their interconnections. It is convenient if the shape geometry is also present since a designer need only indicate the face, edge or vertex at each end of a dimension and the program can compute the nominal size. The designer will normally supply the tolerances, however.

The shape model is regarded as a frame structure whose members correspond to the edges of the object. The members are initially unconstrained in length, and meet at pin joints corresponding to object vertices. For every dimension fixing a distance or angle, or declaring a face to be planar, a corresponding stiffener termed a strut, web or plate, is added to the frame. Struts, webs and plates prevent strain, shear or torsion in parts of the frame, and each gives rise to one or more equations as detailed in [2]. Some dimensions are conventionally not shown on a drawing but have to be inferred: these include struts of zero-length, right-angled webs and plates.

3. Automatic dimensioning

An object which has been described by means of a geometric modeller has enough inherent information to generate its own dimensioning scheme. However, for a modeller that describes only ideal shapes, for example BUILD, the tolerance information is missing and so the dimensioning scheme must necessarily be incomplete. The nature of an engineering drawing is that it shows the salient features of an object in a form which corresponds either to the design or to the manufacture or preferably to both. An automatic dimensioning algorithm will therefore never be able to reconstruct perfectly the designer's thoughts. For these reasons complete automatic dimensioning will always produce inferior results to manual dimensioning.

The ideal strategy is one in which the designer dictates the important dimensions and those with significant tolerances, and leaves the algorithm to note and fill in missing dimensions. The resulting scheme can then be checked for consistency with respect both to itself and to the original description. One method of checking whether or not a scheme is admissible has been referred to above, and while it is an acceptable method for schemes that succeed, for those that fail the test it gives no indication of where extra dimensions may be positioned or of the number required. The time taken by this algorithm is proportional to the cube of the number of vertices; one which improved on this would be preferable for complex objects.

An analysis of the properties of three-dimensional shapes leads to a faster method for checking dimensioning schemes, although this method cannot in general distinguish between good and very good schemes. It does however suggest a method of generating dimensions that emulates the checking, and therefore always yields at least

good schemes. Solid objects can be characterized by two properties: if all the vertices of an object are trihedral, that is they are at the intersection of exactly three faces (and three edges), then the object is simple; if all the faces of an object are bounded by exactly three edges (vertices), then the object is simplicial and is called a deltahedron. Simple objects and deltahedra are topological duals; that is every face or vertex in one has a corresponding vertex or face in the other. Most real objects, however, have neither of these properties. The tetrahedron is the only object which has both properties. Any general object can be converted into either simple or deltahedron form by adding new faces. Most engineering objects tend towards the simple; indeed, all engineering objects may be considered simple because although the designer may think in terms of having four intersecting planes meeting at a point, this condition cannot in general hold true. Four surfaces cut, for example, by a milling machine will always be non-intersecting if measured to a sufficiently fine tolerance. Similarly, by the duality described above, a face of more than three vertices cannot in general be planar even though it may have been so designed.

It can be shown that, under certain conditions, a deltahedron is rigid; hence one way of dimensioning an object is notionally to split each face into triangles and ensure that each triangle is rigid. This method has the disadvantage that, for a simple object, it substitutes struts or webs for plates and so makes the resultant drawing less tidy and, more importantly, does not produce a scheme which models the original design. Another possibility, the dual of the last method, which is suitable for simple objects is to define the lengths of all the edges of an object and to constrain the faces to be planar. For non-prismatic objects, this process defines the solid angles at the vertices thereby making the whole rigid.

An r-fold prism, defined as one whose base is a two-dimensional (r-1)-fold prismatic polygon, has 0,2,3 shear axes and 0,1,3 torsion axes for r = 0,1,2, respectively. A 0-fold prism is a non-prismatic polyhedron; a 2-fold prism is a 3-fold prism. For any axis, the shear can be controlled by fixing the two-dimensional rigidity of a face through which the axis passes and for a torsion axis, it is necessary that at least one adjacent face to any face through which the axis passes is planar. The algorithm which is used to dimension simple objects, those with many parallel faces which are not necessarily prismatic, runs as follows:

(i) the faces are arranged into sets, each set containing a group of parallel faces;

(ii) struts are set up between adjacent faces in a set;

(iii) webs are set up between the first faces of adjacent sets;

(iv) for prismatic objects, the tests on the shear and torsion axes are performed to verify that the object is rigid.

If the total number of items of information provided by the dimensions is not enough then it is necessary to continue by following the algorithm for deltahedron-like objects below. If the object is preprocessed such that any non-trihedral vertices are truncated and replaced by faces (of zero size) then this last step should not be necessary. In (iii), other webs will be implied (the secondary webs of [2]) some of which will be redundant; these will not contribute to the total.

The algorithm for dimensioning objects which can be easily converted to deltahedron form is as follows:

> for each face check that it is rigid in a two-dimensional sense, though not necessarily planar, adding dimensions where necessary to make it rigid.

In practice if all faces are made two-dimensionally rigid except one which is made planar then the object will still be rigid. The views of fig.3 constitute an automatically dimensioned drawing of a die used in a manufacturing process.

4. Dimensioning of curved objects

For curved objects, we proceed as before to set up equations relating shape geometry to dimensions. It is now no longer possible to express the shape entirely in terms of vertex coordinates, as was done for polyhedra since the presence of curved surfaces introduced extra degrees of freedom. In principle, the coefficients of the equations of curves and surfaces could be chosen as the shape variables. However, these do not have an obvious relation to the shape as seen or intended by the designer. Instead we shall make use of quantities such as the radius of an arc or its centre or other information that appears on a drawing. As with polyhedra, we shall again find inferable dimensions: tangency conditions, for example, are seldom shown explicitly but we must still derive equations for the constraints that they apply.

The example in fig.1 (overleaf) shows a two-dimensional shape that contains a curved edge dimensioned in two different ways. The object has eleven degrees of freedom in total, an x and y coordinate at each vertex, plus the radius of the circular arc. After fixing solid body position and rotation, we have eight degrees of freedom to be supplied by the dimensions. In the first case shown in fig.1 (i), the curved edge is defined by its radius and the vertices at its ends. The appropriate unknowns are thus the x and y coordinates at each vertex and the radius. In the second example, we introduce the coordinates of the centre of the arc as additional unknowns and omit one coordinate (x in one case, y in the other) at the vertices at the ends of the arc.

For other kinds of curve and surface, for example B-splines and surfaces, one can imagine schemes in which the defining polygons or polyhedral meshes were dimensioned. In this case it would be natural to use the positions of the polygon or polyhedron vertices as the variables in an analysis of shape. This remains a subject for further investigation.

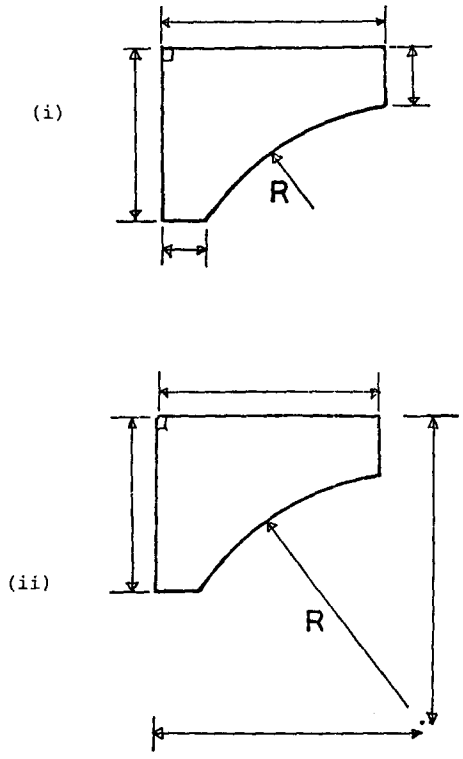


Fig. 1 Dimensioning a curved edge

5. Design alteration

The fact that the constraint equations can be solved to yield the displacements of vertex positions, curves or surfaces for a given set of variations in the dimensions leads to the attractive idea of using the same technique to change the shape of an object through its dimensions. This allows an object to be altered without recourse to a complete re-evaluation of the shape from scratch. However many of the constraint equations will have been derived under the assumption that the variations in the dimensions are small and so in general it will be necessary to employ a relaxation technique to find the consequent displacements. The technique is as follows.

(i) Set the elements of the displacement vector to zero;

(ii) set the variation vector to the difference of the required dimensions and the present dimensions;

(iii) while

- a) construct the rigidity matrix according to the present coordinates;
- b) subtract the product of the rigidity matrix and the displacements from the variations;
- c) solve the equations to find the new displacements;
- d) the product of the norms of the variations and displacements is greater than the finishing criterion

do add the displacements to the vertex coordinates

od

Fig.2 shows an example of the alteration of a two-dimensional figure. The two solid outlines denote, as indicated, the initial shape of the pentagon and its final shape (obtained by increasing one length and decreasing one angle as shown.) A three-dimensional object can be altered in exactly the same way.

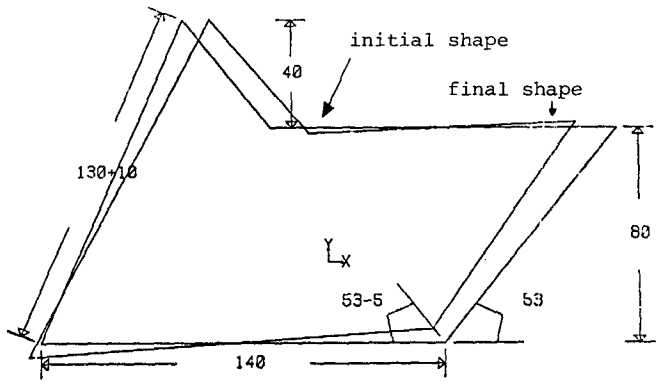


Fig. 2 Pentagon altered by relaxation

The relaxation process just described leads us to consider the possibility of optimising a design according to some criterion ξ and a set of bounds on the independent variables. We require a hill-climbing algorithm of the type used by linear programmers. The hills rest on a plane which is in fact an m-dimensional space: m is the number of independent quantities which we wish to be able to vary, and $m \leq 3v-6$ for an object of v vertices. We can use a geometric modeller to provide procedures to evaluate derivatives of the criterion with respect to the vertex coordinates. We employ the chain rule in matrix form:

$A = B F$ where $a_i = \frac{\partial \xi}{\partial x_j}$ (the derivative with respect to the jth variation),

and $b_i = \frac{\partial \xi}{\partial v_i}$ (the derivative with respect to the ith vertex coordinate),

and $f_{ij} = \frac{\partial v_i}{\partial x_j}$

The optimisation process then is one which tries to find a point where all the $\frac{\partial \xi}{\partial x_i}$ are zero.

Both the relaxation and optimisation techniques must guard against non-convergence and location of minima and local maxima instead of global maxima.

Conclusion

The research described in this paper began as a quest for ways of attaching dimensions and tolerances to shape descriptions. We have found that the problem goes much deeper than that. It concerns, firstly, the avoidance of redundancy in shape descriptions, whether as drawings or as computer data structures. Secondly, it requires that we regard shape descriptions as dynamic mechanisms rather than as static entities. Although our prototype system is as yet under-developed, we can already see how engineering designers will be able to build a nominal shape and derive from that an irredundant description by supplying dimensioning and tolerancing information. They will be able to interrogate the description to find derived geometric quantities and variations and will be able to assess the quality of the irredundant scheme. Designers will surely welcome the facility of changing shapes merely by altering dimensions, and production engineers too will be able to discover the tolerance information needed for the planning of manufacture.

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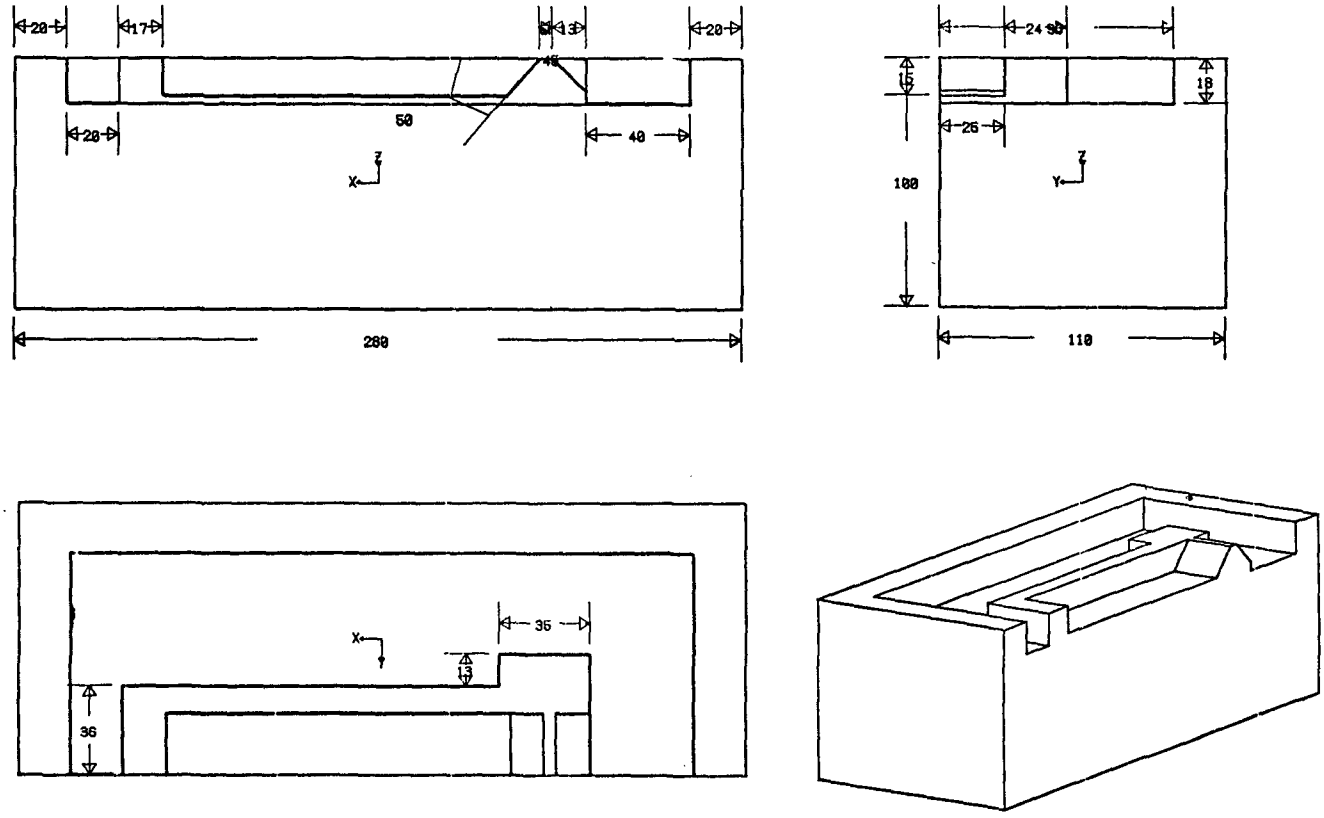


Fig. 3 Automatically dimensioned moulding die